THE COLIMIT CONDITION PRESERVES INJECTIONS OF SET DIAGRAMS BUT NOT OF ABELIAN GROUP DIAGRAMS

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 2586-2666, following the statement at 2565-2584.

Editorial note. Part (1) covers diagrams of sets; part (2) is the counterexample for abelian groups, both inside the same span.

Allowed standard prerequisites (editorial).

- Decomposition into connected components. Lemma lemma-split-into-connected of this chapter, invoked at proof lines 2588-2593. Writes I as a coproduct of nonempty connected index categories satisfying the same condition, reducing (1) to the connected case since colimits over a coproduct decompose.
- Colimits of set-valued diagrams as a quotient description. Section section-limit-sets of this chapter, used without further mention at 2600-2601. The zig-zag chain description of equality in a colimit of sets; the induction runs on chain length.

Source reference. Original lemma 09WS: sources/stacks-categories/pinned-original-source.tex, source lines 2538–2538. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original section 002U: sources/stacks-categories/pinned-original-source.tex, source lines 1807–1807. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let $\mathcal{I}$ be an index category, i.e., a category. Assume that for every solid diagram*

$$\begin{array}{ccc} x & \longrightarrow & y \\ \downarrow & & \vdots \\ z & \dashrightarrow & w \end{array}$$

in $\mathcal{I}$ there exist an object w and dotted arrows making the diagram commute. Then

(1) *an injective morphism $M \to N$ of diagrams of sets over $\mathcal{I}$ gives rise to an injective map $\operatorname{colim} M_i \to \operatorname{colim} N_i$ of sets,*
(2) *in general the same is not the case for diagrams of abelian groups and their colimits.*

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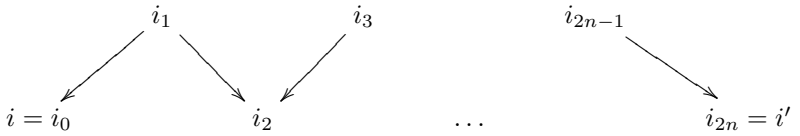
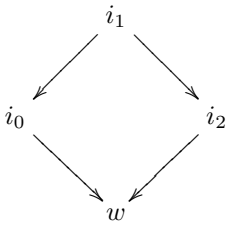
Proof. If $\mathcal{I}$ is the empty category, then the lemma is true. Thus we may assume $\mathcal{I}$ is nonempty. In this case we can write $\mathcal{I} = \coprod \mathcal{I}_j$ where each $\mathcal{I}_j$ is nonempty and satisfies the same property, see Lemma 09WS. Since $\operatorname{colim}_{\mathcal{I}} M = \coprod_j \operatorname{colim}_{\mathcal{I}_j} M|_{\mathcal{I}_j}$ this reduces the proof of (1) to the connected case.

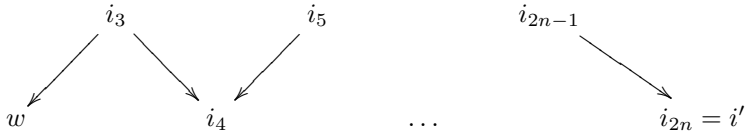
Assume $\mathcal{I}$ is connected and $M \to N$ is injective, i.e., all the maps $M_i \to N_i$ are injective. We identify M_i with the image of $M_i \to N_i$, i.e., we will think of M_i as a subset of N_i. We will use the description of the colimits given in Section 002U without further mention. Let $s, s' \in \operatorname{colim} M_i$ map to the same element of $\operatorname{colim} N_i$. Say s comes from an element m of M_i and s' comes from an element m' of $M_{i'}$. Then we can find a sequence $i = i_0, i_1, \ldots, i_n = i'$ of objects of $\mathcal{I}$ and morphisms



and elements $n_{i_j} \in N_{i_j}$ mapping to each other under the maps $N_{i_{2k-1}} \to N_{i_{2k-2}}$ and $N_{i_{2k-1}} \to N_{i_{2k}}$ induced from the maps in $\mathcal{I}$ above with $n_{i_0} = m$ and $n_{i_{2n}} = m'$. We will prove by induction on n that this implies $s = s'$. The base case $n = 0$ is trivial. Assume $n \geq 1$. Using the assumption on $\mathcal{I}$ we find a commutative diagram



We conclude that m and n_{i_2} map to the same element of N_w because both are the image of the element n_{i_1}. In particular, this element is an element $m'' \in M_w$ which gives rise to the same element as s in $\operatorname{colim} M_i$. Then we find the chain



and the elements n_{i_j} for $j \geq 3$ which has a smaller length than the chain we started with. This proves the induction step and the proof of (1) is complete.

Let G be a group and let $\mathcal{I}$ be the one-object category with endomorphism monoid G. Then $\mathcal{I}$ satisfies the condition stated in the lemma because given $g_1, g_2 \in G$ we can find $h_1, h_2 \in G$ with $h_1 g_1 = h_2 g_2$. A diagram M over $\mathcal{I}$ in Ab is the same thing as an abelian group M with G-action and $\operatorname{colim}_{\mathcal{I}} M$ is the coinvariants M_G of M. Take G the group of order 2 acting trivially on $M = \mathbf{Z}/2\mathbf{Z}$ mapping into the first summand of $N = \mathbf{Z}/2\mathbf{Z} \times \mathbf{Z}/2\mathbf{Z}$ where the nontrivial element of G acts by $(x, y) \mapsto (x + y, y)$. Then $M_G \to N_G$ is zero. $\square$